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Research Article

An Artificial Intelligence Model for the Automatic Classification of the Cervical Vertebral Maturation Stages

Sara AlRashed AlKhlaf, BDS[1,*], Rana AlMurshed, BS, MSc, PhD[2], Maimuna Bashir, BSc, MSc[3], Moshabab Asiry, BDS, MSc[1]

[1]Department of Pediatric Dentistry and Orthodontics, College of Dentistry, King Saud University, Riyadh 11451, Saudi Arabia
[2]Department of Information Technology, College of Computer and Information Sciences, King Saud University, Riyadh 11451, Saudi Arabia
[3]Artificial Intelligence Center of Advanced Studies, Riyadh 12372, Saudi Arabia
*Corresponding author: Sara AlRashed AlKhlaf, BDS, Department of Pediatric Dentistry and Orthodontics, College of Dentistry, King Saud University, King Khalid Road, Riyadh 11451, Saudi Arabia (saradscd@gmail.com).

Abstract

Objective: **To develop an artificial intelligence (AI) model for automatic classification of cervical vertebral maturation (CVM) stages on lateral cephalometric radiographs (LCR).**

Materials and Methods: **Overall, 1140 LCRs were labeled and classified according to the CVM user guide by McNamara and Franchi using a 2-stage pipeline. In the first stage, the LCRs were cropped to extract the region of interest (ROI) using the YOLOv11s.pt model. In the second stage, the extracted ROI were used to train the ResNet-101 deep learning (DL) model to classify CVM stages. Model performance was evaluated using accuracy, recall, precision, and F1-score. Performance comparison between human and AI was conducted by calculating the quadratic weighted kappa and ICC values.**

Results: **Intra- and inter-examiner reliabilities were measured using weighted kappa values of 0.992 and 0.953, respectively. The ResNet-101 DL model showed 84% overall accuracy, and the quadratic weighted kappa and ICC values between human and AI readings were 0.93 and 0.95, respectively.**

Conclusions: **The proposed DL model achieved 84% accuracy in automated CVM stage classification. This AI-based framework enhances objectivity and efficiency of skeletal maturation assessment through standardized and reproducible classification. Accordingly, it reduces the influence of subjective evaluation, particularly where clinical workload is high.**

Advances in Knowledge: **Automated CVM assessment improves objective evaluation of skeletal growth. To our knowledge, this study is the first to integrate YOLOv11-based localization with ResNet-101 for CVM classification, enabling an automated workflow supported by a curated and balanced dataset and a reproducible framework.**

Key words artificial intelligence, cervical vertebrae maturation; deep learning

Introduction

Determining the optimal timing for orthodontic and dentofacial orthopedic interventions is a critical process in clinical decision-making. Timely treatment enables clinicians to leverage the patient's natural growth potential to correct or mitigate jaw discrepancies, enhance facial esthetics, and potentially reduce the need for future surgical intervention. Moreover, it can significantly contribute to improved self-esteem and psychological well-being.[1-3] The assessment of treatment timing can be guided by various biological indicators, including stature, body weight, onset of menarche or voice changes, dental eruption patterns, hand and wrist skeletal development, and cervical vertebral maturation (CVM) assessment.[4-8]

The CVM method was first introduced by Lamparski in 1975,[9] and several indices have subsequently been proposed. It has proven effective and has been used in the orthodontic specialty as it is readily available on the lateral cephalometric radiograph (LCR) taken regularly at the beginning of treatment. Hence, it does not necessitate extra radiation. Despite its widespread use, the CVM method has been criticized for its limited reproducibility, time-consuming nature, and need for expertise. Therefore, recent advancements have focused on automating it using artificial intelligence (AI) to improve consistency and efficiency.[10-13]

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AI is a field of computer science that refers to a machine's ability to perform tasks that require human intelligence.[14] Machine learning (ML) is a subset of AI. ML focuses on creating systems that learn from data, rather than programming using code. Deep learning (DL) is a specialized subset of ML that uses deeper networks resembling neural networks in the human brain. It has also been used for image detection and classification problems.[15]

The application of AI in CVM assessment has evolved rapidly since its initial introduction in 2019.[16] Early studies primarily relied on traditional ML techniques and basic convolutional neural network (CNN) models, demonstrating the feasibility of automated CVM staging.[16,17] More recent work has shifted toward advanced DL architectures, incorporating more sophisticated workflows such as landmark detection and image preprocessing techniques. Some approaches have also explored multi-stage classification strategies and object detection frameworks to improve region-of-interest (ROI) localization and classification performance.[11,18]

Despite these methodological advancements, reported outcomes remain variable, with model performance differing widely across studies due to variations in dataset characteristics, model architecture, and evaluation protocols. This heterogeneity reflects several persistent limitations, including the frequent use of relatively small or single-center datasets, variability in annotation protocols and reference standards, insufficient transparency in preprocessing and augmentation strategies, and lack of clarity in dataset splitting into training, validation, and testing sets. Furthermore, the lack of external validation and direct comparison with expert human performance limits their clinical applicability.[17-20] Consequently, no consensus has yet been established regarding the optimal AI-based approach for CVM staging. Accordingly, the present study was designed to address these challenges by establishing a curated and balanced dataset representing all 6 CVM stages. Using this foundation, we developed and validated an AI model for the detection and classification of CVM stages based on McNamara and Franchi's user guide within a clearly defined and reproducible framework.[12]

Methods

Ethical approval

This study was submitted to and approved by the Institutional Review Board (IRB) of King Saud University (KSU), Medical City (Approval No. E-24-8514). The research was also submitted to and registered at the College of Dentistry Research Centre (CDRC). All experiments were performed in accordance with relevant guidelines and ethical regulations, and all images were anonymized to ensure patient confidentiality.

Data collection

The total sample size in the study included 1140 LCRs. These were randomly divided into 2 separate sets without overlap: 1020 for training and validation, and 120 as an external test set. Both sets were class-balanced and involved both genders (Table 1). The LCRs were obtained from the orthodontic records of patients who visited the orthodontic clinic at the University Dental Hospital, KSU, Riyadh, Saudi Arabia, for orthodontic treatment.

Table 1 Gender distribution among the 6 CVM stages of the training dataset.

Stage	CS1	CS2	CS3	CS4	CS5	CS6
Gender						
Female	74	77	78	117	112	98
Male	96	93	92	53	58	72
Total	170	170	170	170	170	170

All LCRs were taken using a Planmeca ProMax device (Planmeca, Helsinki, Finland), with the following exposure parameters: 68 kV and 5 mA for 18.722 s. The original size was 1939 × 2091 pixels, with a pixel size of 144 μm, and 16 bits per pixel.

The study included LCRs that satisfied the following criteria: (1) patients aged 6-9 years; (2) patients with complete medical records and LCRs demonstrating satisfactory clarity and visibility of the cervical vertebrae (CV): CV2, CV3, and CV4; (3) absence of congenital syndromes and metabolic diseases; and (4) history of special drugs that affect growth and development. To facilitate the transmission of the LCRs to the labeling tool, the images were converted to the Joint Photographic Experts Group (JPEG) file format.

Data labeling

The LCRs were uploaded to an open-source labeling tool called Label Studio (https://ls.crowdata.us). Then, 2 examiners (an orthodontist with 19 years of experience and a fifth-year orthodontic resident from the same institution) independently evaluated the images and classified them into 6 CVM stages (Table 2): cervical stage 1 (CS1), cervical stage 2 (CS2), cervical stage 3 (CS3), cervical stage 4 (CS4), cervical stage 5 (CS5), and cervical stage 6 (CS6) according to the guide proposed by McNamara and Franchi.[12] In addition, bounding boxes were manually defined to identify the ROI, which encompasses CV2, CV3, and CV4. To assess reliability, a subset of 30 images was randomly selected and reevaluated. One examiner repeated the classification after a 2-week interval to determine intra-examiner reliability, while the second examiner independently re-classified the same images to assess inter-examiner reliability. Following the reliability assessment, any disagreements in classification were resolved through consensus with a third examiner with more than 10 years of experience, and the finalized labels were used for subsequent model training. To minimize examiner fatigue, a maximum of 10 LCRs were evaluated per session with optimal adjustment of screen light.

Data cropping and resizing

Using a YOLOv11s.pt model, the ROI were detected and cropped, and the LCRs were resized to 224 × 224 × 3 (Figure 1). As a single-stage detection architecture, YOLO predicts bounding boxes and class probabilities in a single forward pass, enabling efficient and accurate localization of the cervical vertebrae region (CV2-CV4) in LCRs. YOLO-based architecture was selected due to its real-time efficiency and robust performance in object detection, as well as its effectiveness in localizing clinically relevant structures in medical imaging applications.[21]

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Table 2 Characteristics of the CVM stages.[12]

Stage	Characteristics
CS1	The inferior borders of vertebral bodies CV2-CV4 are flat, or sometimes slightly convex. The third and fourth cervical bodies are trapezoidal in morphology. With the posterior border of the vertebral body taller than the anterior border, and the superior surface sloping downward and forward.
CS2	A visible notch along the inferior border of the second cervical vertebra (odontoid process). The lower borders of the third and fourth vertebral bodies remain flat. Both CV3 and CV4 retain a trapezoidal shape.
CS3	Visible notching of the inferior borders of CV2 and CV3; the inferior border of CV4 remains flat. Most of the CV3 and CV4 bodies still retain a trapezoidal shape. In some instances, however, either CV3 or CV4 has a more rectangular horizontal shape.
CS4	All 3 bodies have obvious concavities along their inferior surfaces, so the more important factor now is the shape of CV3 and CV4. At CVS 4, both vertebral bodies have a rectangular horizontal rather than a trapezoidal shape.
CS5	The bodies of CV3 and CV4 become square in shape. At least one of the bodies of CV3 and CV4 is square. If not square, the body of the other cervical vertebra is rectangular horizontal. All 3 cervical bodies have notches, so the presence of notching no longer is important in the differential diagnosis.
CS6	At least one of the third and fourth cervical bodies has assumed a rectangular vertical morphology, with the length of the posterior border being longer than the inferior border. If not rectangular vertical, the body of the other cervical vertebra is squared. In addition, the cortical bone appears better delineated in CS6 than at CS5.

Data augmentation

Data augmentation techniques were applied to the entire set of LCRs to increase the diversity and robustness of the training data. These included horizontal and vertical flipping (along the X and Y axes) as well as rotation by 180°. An initial subset of 760 LCRs was used to evaluate the effect of contrast-limited adaptive histogram equalization (CLAHE) and to optimize preprocessing parameters prior to full-scale implementation. Subsequently, CLAHE augmentation was applied to the entire dataset. The augmented images were then combined with the original dataset, resulting in an expanded dataset of 6120 LCRs in preparation for model training.

Data splitting

The 6120 LCRs were divided into a training set and a validation set comprising 80% and 20%, respectively. The 2 sets were randomly divided. Additionally, to avoid class imbalance, both datasets were constructed to include an equal number of images from each of the 6 CVM stages. The training images were used to train the AI model, whereas the validation set was employed to refine the model's hyperparameters and evaluate the performance of the model.

Data training

Training was performed using the ResNet-101 model with ImageNet-pretrained weights (ResNet101_imagenet1k_v1) (Table 3). ResNet-101 is a deep residual network consisting of 101 layers, designed to mitigate the vanishing gradient problem through residual connections, thereby enabling effective feature learning in deep architectures. ResNet-101 was selected due to its strong feature extraction capabilities and its proven effectiveness in medical image classification tasks, particularly in capturing subtle anatomical features.[22] The model was adapted by replacing the final fully connected layer to classify 6 CVM stages. A 2-stage training strategy was applied, in which the classification head was initially trained with the backbone frozen, followed by fine-tuning of deeper layers to optimize feature extraction for the target dataset. Additionally, data augmentation techniques were applied using transforms.Compose, including color jitter and horizontal flipping. To improve generalization, a dropout layer with a rate of 0.5 was added before the final classification layer. The AdamW optimizer, which is an Adam optimizer with weight decay regularization, was also used. In addition, a ReduceLROnPlateau scheduler was employed to facilitate optimal model convergence and prevent performance stagnation during training.

The following metrics were used to evaluate the performance of the AI model: accuracy, recall, precision, and F1-score. Accuracy is a metric used to describe the performance of a model and is calculated as the ratio of the number of correct predictions to the total number of predictions. Recall (true positive rate) refers to the probability of a positive test, conditioned on truly being positive. Precision (true negative rate) refers to the probability of a negative test, conditioned on truly being negative. F1-score harmoniously weighs recall and precision to completely evaluate the AI performance. These metrics are defined as follows:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

$$\text{F1} - \text{score} = 2 \times \frac{\text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}}$$

Where TP, TN, FP, and FN are the numbers of true-positive, true-negative, false-positive, and false-negative samples, respectively.

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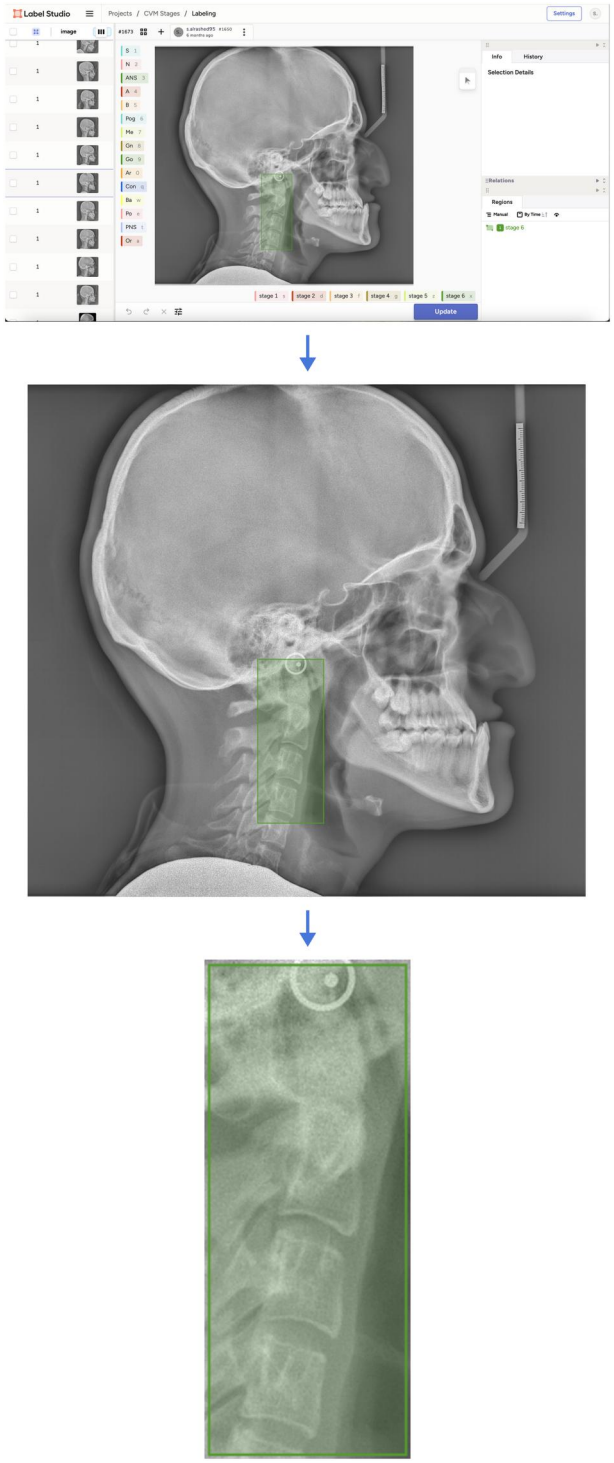


Figure 1 Example of labeling the actual CVM stages on the LCR using Label Studio, with an example of the cropped ROI.

All statistical analyses were performed using the Statistical Package for the Social Sciences (SPSS) version 21.0 (SPSS 21.0 Software Package, Armonk, NY, USA). The weighted kappa (κ_w) and percent agreement were calculated to measure the intra- and inter-reliability of the examiners. To compare the performance of the human examiner to that of the AI model, an external test set consisting of 120 LCRs was independently labeled and classified by both. Results were compared using weighted kappa (quadratic) and intraclass correlation coefficient (ICC 3,1).

Results

Intra- and inter-examiner reliability

The intra-examiner reliability showed a weighted kappa value of 0.992, indicating an almost perfect agreement. Additionally, the percentage agreement was 96.67%, indicating high reliability. Both metrics confirmed excellent consistency in the CVM staging. Moreover, the inter-examiner reliability showed a weighted kappa value of 0.953, indicating an excellent level of agreement. Percent agreement was 90%, corroborating the almost perfect agreement suggested by the weighted kappa values.

AI model evaluation

The accuracy, recall, precision, and F1-score values for ResNet-101's DL model were all calculated using the following confusion matrix (Figure 2). The overall performance of the model was evaluated across 6 CVM stages and achieved an overall accuracy of 84% (Table 4). To provide a more comprehensive and balanced evaluation, the macro-averaged and weighted-averaged metrics were calculated, and both were 84%. The model demonstrated strong performance, with a macro-averaged precision, recall, and F1-score of 0.84 for each metric. This indicates a robust and consistent ability of the model to classify images across all 6 stages. For a detailed stage-by-stage analysis, the values along the main diagonal are high for each of the 6 CVM stages, representing the number of correctly classified images. CS6 demonstrated an F1-score of 0.91, a recall of 0.93, and a precision of 0.89. CS5 and CS4 both achieved an F1-score of 0.87, with recall rates of 0.83 and 0.81 and precision rates of 0.91 and 0.93, respectively. CS3 and CS2 showed similar F1-scores; 0.78 for CS3 and 0.77 for CS2. Their recall rates were 0.83 and 0.79, and their precision rates were 0.73 and 0.74, respectively. CS1 demonstrated an F1-score of 0.83, with a recall of 0.82, and a precision of 0.84. The overall performance across the 6 CVM stages is further illustrated in Figure 3, which presents a bar chart depicting performance metrics across the 6 CVM stages. In terms of agreement, comparison between the human examiner and the AI model yielded a quadratic weighted kappa value of 0.93 and an ICC (3, 1) value of 0.95. Moreover, Figure 4 presents

Table 3 Properties of the pretrained ResNet101.imagenet1k_v1 DL model.

Network model	Depth	Size (MB)	Parameters (millions)	Input image size
ResNet-101 (ours)	101	~167 MB	~44.6 M	224 × 224 × 3

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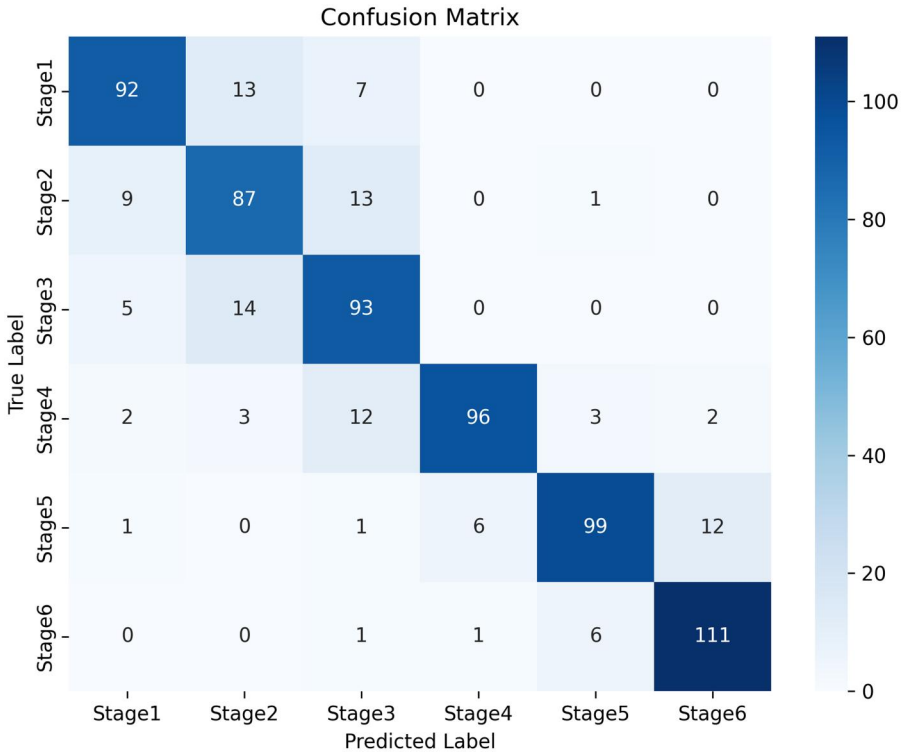


Figure 2 Confusion matrix of the training set across the 6 CVM stages.

Table 4 Evaluation of the performance of the DL model.

Performance Metric CS	Precision	Recall	F1-score
CS1	0.84	0.82	0.83
CS2	0.74	0.79	0.77
CS3	0.73	0.83	0.78
CS4	0.93	0.81	0.87
CS5	0.91	0.83	0.87
CS6	0.89	0.93	0.91
Accuracy			0.84
Macro average	0.84	0.84	0.84
Weighted average	0.84	0.84	0.84

gradient-weighted class activation mapping (Grad-CAM), an interpretability technique used to visualize the regions of an image that contribute most significantly to the model's predictions. This method operates by computing the gradients of the predicted class with respect to the feature maps of the final convolutional layer, generating a heatmap that highlights the most influential regions.

Discussion

The application of AI to automate diagnostic processes in orthodontics offers promising solutions to traditional clinical challenges. Although a multitude of AI models have been proposed in the literature for CVM assessment, no standard method has been established or widely accepted to date.[23,24] The first application of AI in CVM assessment was reported by Kök et al.,[16] who utilized traditional ML techniques, after which the field began to transition toward DL approaches. This shift was pioneered by Makaremi et al.[11] and is attributed to the superior capability of DL in capturing complex image features, leading to better performance. Several DL architectures have been explored in this domain, including VGG, DenseNet, and ResNet.[23] ResNet is among the most efficient DL models that have achieved state-of-the-art accuracy on large-scale image datasets such as ImageNet. It is uniquely deep, yet not overly complex, and has been used in medical imaging for detection and classification.[22] Moreover, there are multiple variants of ResNet that differ in depth and feature-learning capacity, allowing their adaptation to tasks of varying complexity.[22] In the present study, ResNet-101 was customized through transfer learning (TL) and tailored for CVM stage classification using a curated dataset. TL is an approach that transfers the knowledge a model gained from training on a large dataset to be utilized for another similar task.[24] This is valuable in medical imaging because collecting large amounts of labeled data is often challenging.[25] However, TL only began to emerge in the classification of CVM staging in recent publications from 2022 onward.[26]

YOLO is a real-time object detection algorithm distinguished by its ability to balance accuracy and speed. Since its introduction, YOLO architecture has undergone successive iterations, with each version building upon its predecessors to address prior limitations and achieve improved detection performance.[27,28] The latest formally documented iteration in the YOLO family is YOLOv11. In the present study, as part of preprocessing prior to model training, the YOLOv11s.pt model was used for ROI cropping and image resizing. To our knowledge, this is the first report on the use of YOLOv11. The cropping step

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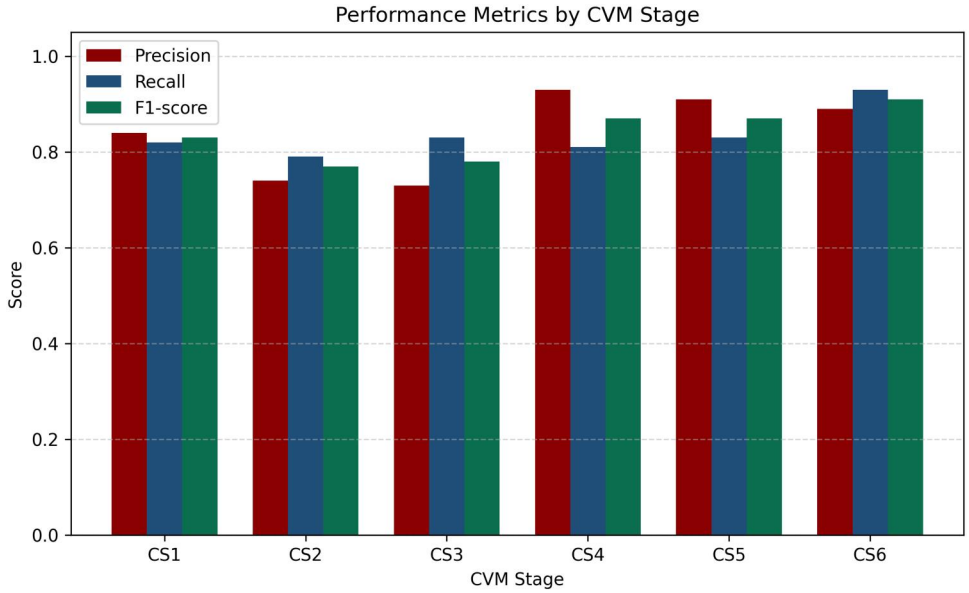


Figure 3 Bar chart depicting the precision, recall, and F1-score metrics per stage, highlighting the classification performance across the 6 CVM stages.

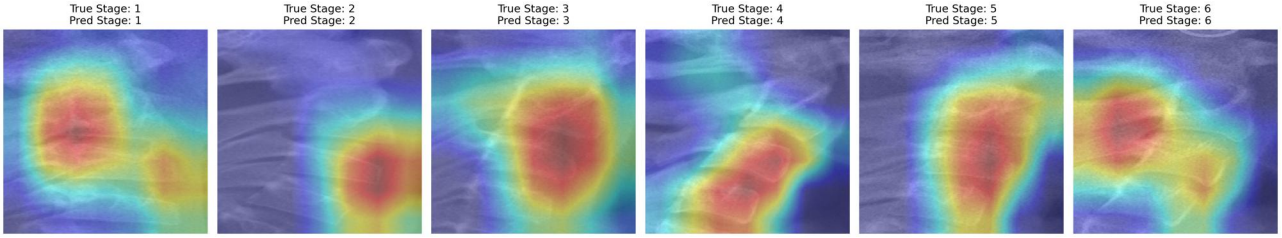


Figure 4 Gradient-weighted class activation mapping (Grad-CAM).

has been documented to enhance the performance of trained DL models in classifying the CVM stages.[11,24,29]

In the present study, a 2-stage DL framework was adopted, in which YOLOv11s was utilized for ROI localization, followed by ResNet-101 for classification. Although YOLO-based architectures are capable of performing simultaneous detection and classification, the separation of these tasks was intentionally implemented to enhance performance for CVM classification. Specifically, CVM staging relies on subtle morphological variations of the CV, which necessitate focused and high-capacity feature extraction. By first localizing and cropping the ROI, the classification model was trained exclusively on relevant anatomical structures, thereby reducing background noise and improving discriminative performance. Moreover, this 2-stage design enabled the application of data augmentation techniques after ROI cropping, contributing to improved robustness and model generalization. Collectively, this 2-stage framework allows each model to specialize in its respective task, resulting in a more controlled and effective approach for automated CVM classification.

Consequently, data augmentation techniques were implemented in this study to enhance the diversity of the training data and improve the model's robustness. It has been documented that augmentation enables the AI model to extract a richer set of features from the training data. However, the application of data augmentation in medical imaging requires careful consideration. This is because anatomical features are highly structured, and introducing artificial variability can distort critical features and affect model reliability.[23] Therefore, in the current study, augmentation techniques were applied cautiously. They were limited to transformations that preserved anatomical proportions and spatial features with no alteration in image size, ratio, or dimensional geometry.

After augmentation, the data were divided into training and validation sets of 80% and 20%, respectively. Because of the limited sample size, most images were used for model training and validation to preserve the training signal. Additionally, an external test set of 120 images served for evaluation to compare the performance of the AI model with that of the human examiner and to assess generalizability and clinical relevance.[30,31]

In the current study, the model attained an overall accuracy of 84% on the validation set. The reported accuracy primarily represents the ResNet-101 DL model's performance after the cropped images by YOLO were manually reviewed, to ensure accurate localization prior to classification. This performance surpasses earlier models, such as that of Mohammad-Rahimi et al.,[26] who reported a validation accuracy of 62.63% in a 6-stage CVM classification and 75.76% in a 3-stage CVM classification with ResNet-101 as their best-performing model. It also exceeds the performances reported by Atici et al.[24,32] Their

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custom CNN performed better than MobileNetV2, ResNet-101, and Xception, and achieved a 75.11% validation accuracy in 6-stage CVM classification. Another study used a novel continuous classification system for CVM staging employing TriPodNet and achieved validation accuracies of 81.17% and 75.96% for female and male patients, respectively. Moreover, the accuracy of this study model was higher than that of the multiple DL models used by Li et al.,[23] such as ResNet-152, DenseNet161, GoogLeNet, and VGG16, with 67.32%, 66.22%, 63.37%, and 63.04% validation accuracies, respectively. Furthermore, a recent study by Motie et al.[20] used ResNet-101 classification models and reported cross-validation accuracies of 82.70 ± 3.59 in classifying CS1-CS3, and 84.68 ± 3.57 in classifying CS4-CS6 on a dataset comprising 2325 LCRs. Kim et al.[33] reported a cross-validation accuracy of 62.5% using 5-fold cross-validation. Although both metrics—validation accuracy and cross-validation accuracy—reflect model performance during development, they are not the same. The former is derived from a fixed validation subset, whereas the latter is obtained through repeated train-test resampling and can vary in the number of cross-validation folds. Therefore, comparisons between the 2 metrics should be interpreted cautiously and only serve to provide a contextual benchmark rather than a direct performance comparison.[19,33] Considerable variability was reflected in the reported accuracies throughout the literature. This wide range of variability can be attributed to multiple interrelated factors.[18,19] CVM represents a continuous and subjective process that is discretized into ordinal stages, introducing label ambiguity. In addition, studies employed heterogeneous AI approaches ranging from traditional ML approaches to DL models with varying architectures, depths, and complexities.[17] Performance metrics are further confounded by the use of different evaluation metrics, data splitting, and stage classification categorization.[34,35] Additionally, differences in dataset characteristics, including sample size, image quality, population composition, and class imbalances across stages, may play a role.[17] Lastly, methodological choices, such as the application of data augmentation or the use or omission of ROI segmentation, also influence model behavior and reported outcomes.[26]

The results of this study demonstrate that the trained model effectively classified all 6 CVM stages with high overall accuracy and strong F1-scores. Although a minor confusion was observed, the model's performance remained consistent and robust. The confusion matrix raises an important point: greater confusion is evident between consecutive stages. Notably, approximately 78.6% of all misclassifications occurred between adjacent stages, indicating that most errors were confined to transitional boundaries rather than substantial diagnostic discrepancies. This pattern is particularly evident between CS2 and CS3, with 14 instances of CS3 classified as CS2. Similarly, 13 instances of CS2 were classified as CS3. This reflects the subtle differences between stages and the continuous nature of the CVM process, where stages may blend into each other, creating borderline cases where a patient might be in late CS2 and early CS3. A similar pattern of confusion between CS2 and CS3 has been reported in prior studies.[23,33] Importantly, these stages correspond to the transition into the pubertal growth phase, with CS3 commonly associated with the onset of the mandibular growth spurt. Therefore, misclassification at this stage may influence the timing of growth-modification treatment. However, given that these are adjacent stages within a similar growth window, such discrepancies may be considered clinically less critical than errors involving non-adjacent stages. Moreover, CVM staging is typically interpreted alongside other clinical indicators, which may mitigate the impact of these borderline misclassifications and support the use of AI as an adjunctive, rather than standalone, clinical tool. In addition, late-maturation CS6 and CS5 showed the highest detection accuracy in this study, reflecting their distinct vertebral morphology. The superior detectability of CS6 has been similarly reported in earlier findings in the literature.[10,36] Furthermore, the lowest detection accuracies in the current study can be seen in CS1 and CS2. This finding is in accordance with previous research that has reported ambiguity between CS1 and CS2 due to subtle morphological features. Hence, several studies have joined both in a single stage, called the prepubertal stage.[24,26,34] In this study, the model demonstrated high detection reliability for CS3 and CS4. These 2 consecutive stages coincide with the period of maximal growth velocity and are therefore clinically critical for treatment timing. Overall, these findings indicate that the model maintains consistent performance across stages while preserving the ordinal structure of CVM classification. Additionally, the dataset in the current study is perfectly balanced, with each stage containing an equal number of images. Hence, the macro-averaged and weighted-averaged metrics produced exactly the same numerical result of 84%.

A key strength of this study was the development of a custom and balanced dataset. It included multiple ethnicities, the majority of whom were Saudis, and both genders. In addition, to enhance generalizability, the dataset contained LCRs with common imaging artifacts (e.g., earrings, piercings, and different hair accessories), orthodontic appliances (e.g., brackets, retainers, and TADs), surgical plates and screws, as well as LCRs with varying image quality. Moreover, the data splitting was clear, with no data leakage between the subsets. Additionally, a human-AI comparison was performed and reported numerically, which signified almost perfect agreement and excellent reliability. This suggests that the customized model performed comparably to a human expert, offering the clinical practice a reliable and efficient tool for CVM assessment. One limitation of this study was the use of a single-center dataset derived exclusively from patients treated at the University Dental Hospital, which may limit the generalizability of the model. Additionally, despite being larger than many previously published datasets, the overall sample size remains relatively small for DL applications. To enhance the performance and generalizability of AI models in CVM assessments, future research should focus on expanding dataset size and diversity across multiple centers. Integrating additional variables, such as chronological age and sex, may further improve accuracy. In addition, evaluating a broader range of advanced DL architectures could help optimize performance and strengthen model reliability for clinical deployment. Although AI has demonstrated significant promise and encouraging results, it still requires rigorous validation and ongoing human oversight. Given the complexity of clinical care, AI systems should operate within a human-in-the-loop (HITL) approach. With continuous supervision and refinement, AI has the potential to transform dental practices by enhancing efficiency and reducing costs.[18]

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Conclusions

- This study established a curated and balanced dataset of LCRs representing all 6 CVM stages, providing a robust foundation for automated skeletal maturation assessment.
- A novel 2-stage framework combining YOLOv11-based localization with ResNet-101 classification achieved 84% overall accuracy in an automated 6-stage CVM assessment, demonstrating strong performance in a clinically relevant multiclass task.
- The results support the role of AI as a decision-support tool that enhances the efficiency and standardization of CVM assessment, aiding clinical decision-making in routine practice.

Conflicts of interest

The authors declare no conflicts of interest.

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